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Network flow
Graded Assignment • 30 min

Due Mar 31, 11:59 PM IST

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

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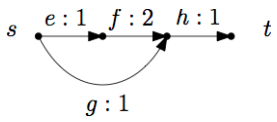
1. Recall that a flow network is a directed graph $G = (V, E)$ with a source s, a sink t, and a capacity function $c : V \times V \to \mathbb{R}_0^+$ that is positive on E and 0 outside E. We only consider finite graphs here. Also, note that every flow network has a maximum flow. This sounds obvious but requires a proof (and we did not prove it in the video lecture). 1 / 1 point

Which of the following statements are true for all flow networks (G, s, t, c)?

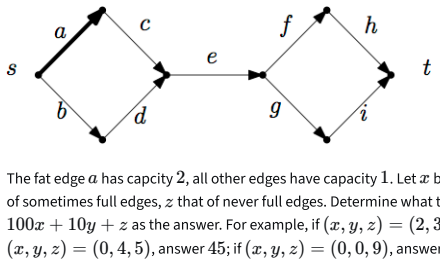
Nice work

2. Let (G, s, t, c) be a flow network, $G = (V, E)$. A directed edge $e = (u, v)$ is called *always full* if $f(e) = c(e)$ for all maximum flows f; it is called *sometimes full* if $f(e) = c(e)$ for some but not all maximum flows; it is called *never full* if $f(e) < c(e)$ for all maximum flows. 1 / 1 point

Let $(S, V \setminus S$ be a cut. That is, $s \in S, t \in V \setminus S$. We say the edge $e = (u, v)$ is crossing the cut if $u \in S$ and $v \in V \setminus S$. We say e is *always crossing* if it crosses every minimum cut; *sometimes crossing* if it crosses some, but not all minimum cuts; *never crossing* if it crosses no minimum cut. For example, look at this flow network:



The edges e, g are sometimes full and never crossing; f is never full and never crossing; h is always full and always crossing. Alright, now it's your turn. Consider this network:



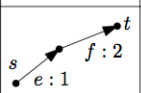
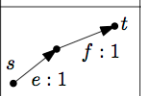
The fat edge a has capcity 2, all other edges have capacity 1. Let x be the number of always full edges, y that of sometimes full edges, z that of never full edges. Determine what these numbers are and enter $100x + 10y + z$ as the answer. For example, if $(x, y, z) = (2, 3, 4)$, answer 234; if $(x, y, z) = (0, 4, 5)$, answer 45; if $(x, y, z) = (0, 0, 9)$, answer 9.

Nice work

3. Again refer to the above flow network on eight vertices. Now let x be the number of always crossing edges in the above picture, y that of sometimes crossing edges, and z that of never crossing edges. Again, compute $100x + 10y + z$ and enter it as your answer. 1 / 1 point

Nice work

4. An edge e can be (x) always full, (y) sometimes full, (z) never full; it can be (x') always crossing, (y') sometimes crossing, (z') never crossing. So there are nine possible combinations: (xx') always full and always crossing, (xy') always full and sometimes crossing, and so on. Or are there? Maybe some possibilities are impossible. Let's draw a table: 1 / 1 point

The edge e is:	always full	sometimes full	never full
always crossing		Possible or impossible?	Possible or impossible?
sometimes crossing		Possible or impossible?	Possible or impossible?
never crossing	Possible or impossible?	Possible or impossible?	Possible or impossible?

Take a piece of paper, complete the 3×3 table below by either finding an example illustrating that the combination is possible or by concluding that it is impossible. To do so, you will need the Max Flow Min Cut Theorem plus a bit of your own thinking. Once you have completed the table, select which of the following statements are true: There is a flow network and an edge e that is...

Nice work